

TESTING OF HYPOTHESIS

Module 5 & 6 Tutorial sheet BMAT202L-PROBABILITY & STATISTICS

- 1) A soft drink company claims the average filling is 500 ml. The population standard deviation is 8 ml. A random sample of 64 bottles shows a mean of 497 ml. Test at 5% significance level whether the company is underfilling.
- 2) Two machines produce bolts.

Machine	Sample Size	Mean Length	σ
A	50	10.2 cm	0.5
B	60	10.0 cm	0.6

Test at 1% level whether the machines produce different mean lengths.

- 3) A university claims that 60% students pass in first class. In a sample of 200 students, 104 passed first class. Test at 5% level whether the claim is true.
- 4) A manufacturer claims variance $\sigma^2 = 25$ and mean = 100. Sample of 100 items has mean 103. At 1% significance, test the claim.
- 5) A population has $\sigma = 12$. A sample of 36 has mean 52. Test $\mu = 50$ at 5%.
- 6) A company claims defect rate is 3%. In 500 items, 25 are defective. Test the claim.
- 7) Two population means differ by 5 units. Given $\sigma_1=4$, $\sigma_2=6$, $n_1=64$, $n_2=81$. Is difference significant at 1%?
- 8) A manufacturer claims the average battery life is 50 hours. A sample of 16 batteries shows:
Sample mean = 47 hours
Sample standard deviation = 6 hours
Test at 5% significance level whether the mean life is less than 50 hours.
- 9) Two teaching methods are compared.

Method	n	Mean	SD
A	10	78	5
B	12	72	6

Test at 5% level if there is a significant difference.

- 10) A manufacturing company claims that the average life of its LED bulbs is 1200 hours. A random sample of 16 bulbs gave the following lifetimes (in hours):
1180, 1215, 1195, 1220, 1175, 1205, 1230, 1160, 1190, 1210, 1185, 1225, 1170, 1200, 1240, 1155
(a) Test at 5% level of significance whether the true mean life differs from 1200 hours.

(b) Construct a 95% confidence interval for the population mean.

- 11) A company introduced a training program. The productivity scores before and after training are given below:

Employee	Before	After
1	65	70
2	72	75
3	68	74
4	70	73
5	66	72
6	74	78
7	69	73
8	71	76

Test at 1% significance level whether training improved productivity.

- 12) Two independent samples:

Sample1:

$n_1 = 12$, mean = 85, SD = 6

Sample2:

$n_2 = 10$, mean = 80, SD = 8

(a) Test equality of variances at 5% level.

(b) Using appropriate t-test, test difference of means at 5% level.

- 13) A random sample of 12 observations from a normal population gives:

Mean=78

Standard deviation = 6

(a) Construct 95% confidence interval for population mean.

(b) Test whether $\mu = 75$ at 5% level.

- 14) In a survey of 160 families, the number of male children was recorded:

No. of Male Children	0	1	2	3	4
No. of Families	18	52	56	28	6

Assuming the probability of a male birth is 0.5, test at 5% level whether the data follow a binomial distribution.

- 15) A survey was conducted to study relationship between smoking and lung disease:

	Lung Disease	No Disease	Total
Smokers	40	60	100
Non-Smokers	10	90	100
Total	50	150	200

Test at 5% level whether smoking and lung disease are independent.

16) A study of 300 individuals gives:

Education Level	Low Income	Middle Income	High Income
School	40	30	10
College	35	50	25
Postgraduate	15	45	50

Test whether education level and income are independent at 5% level.

17) The following observations are drawn from two normal populations:

Sample A: 15, 18, 14, 17, 19, 16, 20, 13

Sample B: 12, 11, 15, 14, 13, 10, 16

Test at 5% level whether the variances of the two populations are equal.

18) Two samples:

Sample1:

$n_1=15$

Mean=120

Variance = 100

Sample2:

$n_2=12$

Mean=110

Variance = 64

Test equality of variances at 5% level.